

Question	Marking Guidance	Mark	Comments
08.1	1. Mutation (in <i>KRAS</i> gene) OR Change in <u>base</u> sequence (of <i>KRAS</i> gene); 2. Change in (signalling) protein OR More (signalling) protein/K-Ras produced; 3. (Results in) rapid/uncontrollable cell division;	3 (3 x AO2)	1. Accept named mutation e.g. substitution. 1. Accept mutation in promoter gene. 1. Ignore epigenetic modifications. 2. Accept change in amino acid sequence (of protein). 2. Reject 'no protein produced' or suggests a non-functional protein is produced. 3. Accept cell division cannot be regulated. 3. Ignore growth. 3. Reject meiosis for cell division, but accept mitosis.

Question	Marking Guidance	Mark	Comments
08.2	1. A/untreated (type II diabetes) lowest survival (time) OR A/untreated (type II diabetes) lowers/reduces survival (time compared with control group); 2. B/metformin increases survival (time) the most OR B/metformin has the highest survival (time); 3. C/combined drugs increases survival (time); 4. Groups A and B have a significant difference (in survival time compared with control); 5. Group C has no significant difference (in survival time compared with control); 6. (In group C) other drugs have reduced effect of metformin OR B/metformin is more effective (treatment/drug) than C/combined drugs;	5 max (5 x AO3)	1 to 6. Accept descriptions of each group e.g. A = no drugs, B = metformin, C = combined drugs. 4 and 5. Reject 'results are significant' or 'results are not significant' once, but only where there is no indication that these results are 'different', 'greater', 'reduced' etc. 4 and 5. Accept 'not due to chance' for 'significant' and converse for 'not significant'. Ignore reference to sample size, repeats or a single study.